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The influence of outdoor thermal comfort on acoustic comfort of urban parks based on plant communities

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ABSTRACT

As human beings live in an environment where various factors interact and integrate, it seems vital to examine the effects of these factors on environmental perceptions. Outdoor thermal and acoustic environments are the main factors affecting the overall comfort of urban spaces. This research uses measurements and questionnaire surveys to explore the effects of outdoor thermal comfort dimensions on acoustic comfort, emphasizing the plant community in a city park (the El Goli Park of Tabriz, Iran). In this regard, the general comfort of the psychological equivalent temperature (PET) under different conditions of the thermal-acoustic environment has been evaluated and predicted. The results showed that a higher thermal sensation vote (TSV) and PET result in a significant decrease in soundscape quality (SQ), overall impression (OI), and the percentage of reported natural dominant sound (%RNDS), while an inverse trend has been reported in thermal comfort vote (TCV). On the other hand, by increasing TSV and PET, reported loudness (RL) has recorded an upward trend. However, this process was completely reversed in TCV. Furthermore, the analyzes showed that PET has a lower effectiveness compared to TSV and TCV and, it should be stated that the three indicators of thermal comfort did not show any significant relationship with landscape quality (LQ) and sound pressure level (SPL). These findings are important for managing and improving outdoor acoustic comfort and soundscapes, and it is expected that future studies will consider the acquired results in various green urban environments and provide a practical framework for landscape architects and urban planners.

1. Introduction

Indiscriminate urbanization as one the phenomenon of the present century leads to a variety of environmental problems, including water shortage [1], industrial and community waster [2], land pollution [3], increased density of population (S. [4], noise discomfort [5], and climate change [6]. In this regard, exposure to noise in outdoor environments has numerous effects on human health, such as hearing loss, sleep problems, cardiovascular and mental health disorders [7,8]. On the other hand, the direct impact of climate changes on people's quality of life and the creation of a local phenomenon called the urban heat island (UHI) has led to thermal discomfort in urban spaces [9].

Urban parks are recognized as an integral component of the intricate urban ecosystem network, which provides various advantages to urban residents [10]. Open green areas in urban parks can improve mental and physical health by decreasing the exposure of inhabitants to air pollutants, noise pollution, and extreme heat as well as offering psychological relaxation and stress relief [11]. Plant Covers have been reported as one of the most important factors determining the quality of urban spaces in the acoustic comfort of urban spaces [12–15],followed by thermal comfort [16–18].

Acoustic comfort is the basic feeling of users towards the acoustic environment [19]. Studies show that acoustic comfort is beyond noise control and includes subjective interpretations of a sound in a particular context [20]; F [21]. The term "soundscape" has been defined as the "acoustic environment as perceived or experienced and/or understood by a person or people, in context" [22]. Typical studies dealt with soundscape experiences such as sound level, perceived loudness [23], perceived dominant sound [24], and pleasantness for sounds [25] etc. These experiences may be influenced by different sound sources [26,27], the social/demographic/behavioral characteristics [28], landscape features [29] and physical factors such as temperature, humidity and

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sunlight (W [30,31]. It is important to combine a variety of quantitative and qualitative research methods to measure the objective acoustic environment (using acoustic measurements, noise maps, etc.) and subjective acoustic perception using interviews and questionnaires [32,33]; M [34].

On the other hand, thermal comfort can be expressed as the physiological interval in which the human able to operate or tolerate the environment [35]. It is the condition in which the human body feels satisfied with the thermal environment [36] and is affected by heat conduction, convection, radiation, and loss of evaporative heat [37]. Several factors such as air temperature, humidity, air flow, solar radiation, terrestrial radiation, metabolic heat, and clothing insulation are factors and the energy budget of a person in outdoor spaces [38,39]. In the evaluation of the outdoor thermal environment, environment microclimatic condition, physiological components (such as the metabolism rate), and behavioral elements are essential for better thermal comfort [40].

It is necessary to evaluate and identify different aspects of thermal comfort as one of the most important environmental factors affecting soundscapes [14]; M [34].. To evaluate acoustic comfort conditions, only understanding the acoustic factors is not enough, and multiple environmental stimuli, including thermal comfort, visual beauty of the environment, and greenness can be effective in the perceived quality of the acoustic environment. Studies showed that thermal conditions significantly affected feelings of loudness or calmness of sound [41–43]. Temperature changes have also been shown to increase or decrease perceived acoustic comfort [44]. Meanwhile [45], suggested that the thermal environment had no effect on the measured noise levels. Although the stated results can have different effects in different spatial conditions, including the variety and extent of natural elements and different socio-cultural factors such as age, it is predicted that the acoustic environment perceived in outdoor spaces is related to the experienced thermal comfort conditions.

Plants are one of the influential indexes in managing environmental conditions that can lead to better outdoor comfort conditions. These include an improved quality of visual comfort [46] and thermal comfort [47,48]; J [49]. In terms of acoustic comfort evaluation, previous studies pointed out the importance of plants in mitigating noise pollution by plants, tree [50–52], or by land use parameters emphasizing green spaces and their ratio [53] [54], or the psychological and physiological impacts of noise reduction [55]; F [56]. Some research also found that non-acoustical features including plants can increase acoustic comfort quality by producing natural soundscapes such as tree rustling [57–59].

At the same order to regulate outdoor thermal comfort, solar attenuation, and evaporative cooling capacity can provide relatively suitable thermal comfort conditions in different cities [60]. These effects are compared to the reduction of air and surface temperature under the tree canopy compared to other areas [61,62]. Shade in trees is also considered as one of the determining factors in improving thermal comfort to cope with the heat sensation of the day and reduce the mean radiant temperature (MRT). Numerous researches have analyzed the impact of trees as a cooling strategy on urban climate and regulation of human thermal comfort in Asia [63]; H. [18,36,64,65]; L. [66], Europa [67–71] and North America [72]; Y [73]. which points to the undeniable effect of urban trees. The summary of the studies on the influence of plant spaces on outdoor thermal comfort and acoustic comfort, are highlighted in Table 1.

The review of previous research illustrated that the role of plant covers in improving acoustic and thermal comfort has been well established. But few studies have examined the relationship between these two indicators in the outdoor environment with an emphasis on natural elements, among which the role of the effect of acoustic comfort on thermal comfort has been further analyzed [14,44,77–79] (Table 1).

This study focuses on the combined effects of thermal comfort on the acoustic environment, with an emphasis on changes in the plant community of El Goli park, Tabriz city, Iran. In this regard, first, the acoustic

Table 1
Summary of research on the acoustic and thermal comfort of plant spaces and combined studies of both.

Reference	City	Method	Suggestion
[51]	Antwerp	Measurement noise data	Trees-covered areas had slightly lower noise levels than the grass.
[52]	Wolfratshausen	Measurement noise data	Further noise reduction is associated with an increase in the ground effect.
[50]	London	Measurement noise data	Green roofs offered significant noise abatement over 1 kHz in the courtyard, while the vegetated facade was effective at reducing noise levels at low frequencies.
[74]	–	Review	The quality of a "place" is strongly influenced by our perception of sound in the surrounding environment.
[53]	Eight UK cities	Measurement noise data	On a micro-scale, different areas can have the same building coverage but different noise levels.
[56]	Hangzhou	Measurement noise data	The results suggest that the emotions aroused by noise and visual stimuli manifest themselves in the synchronization of the beta frequency band and the desynchronization of the alpha frequency band, indicating that landscape plants can moderate or buffer the effects of noise.
[55]	Plovdiv	Measurement noise data	The effects on annoyance were mediated by noise sensitivity.
[54]	Bradford	Measurement noise data	The percentage of people feeling more relaxed after visiting the spaces was closely related to overall assessments of perceived tranquility.
[59]	Holland	Review	Discussions are presented about the methods used in soundscape research with a special focus on human perception assessment.
[57]	Rome	Field survey	Even if the level of sound pressure is nearly always higher than the limits commonly used to define quiet areas, the sound environment in urban parks is often considered ‘good’ or “excellent”.
[58]	Jinan	Measurement noise data	Soundscapes vary with time of day and season and are influenced by the surrounding environment and human behavior.
[63]	Tehran	Simulation	Air quality rates correlate significantly with green spaces and wind direction.
[64]	Singapore	Field measurements	Trees with large canopies, either in diameter or height, are most effective in improving the mean PET at precinct level.
[18]	Roorkee	Field measurements and simulation	Fraxinus excelsior can produce the most effective conditions for thermal comfort in Roorkee.
[36]	Isfahan		Trees with high trunks and wide, overlapping crowns

(continued on next page)

Table 1 (*continued*)

Reference	City	Method	Suggestion
		Field measurements and simulation	were found to be the most suitable forms of greenery to reduce PET levels during the day in Isfahan.
[65]	Chongqing	Field measurements	Thermal comfort of green spaces is one of the most important factors that affect people's activities.
[66]	Wuhan	Field measurements and simulation	Evergreen species with the height-to-distance ratio of trees <2 and all trees with a large LAI and canopy diameter have been proposed.
[75]			
[67]	Sevilla and Madrid	Field Measurements and Simulation	wider semi-open spaces and relatively suitable vegetation bring more favorable conditions for visitors.
[68]	Lisbon	Field measurements	Trees are capable of a maximum Ta reduction up to 6.9 °C.
[69]	Trapani	Monitoring	Green surface temperature decreases by up to 20 °C.
[70]	Athens	Field measurements	Trees are considered influential factors in controlling climatic conditions.
[71]	Erzurum	Field measurements	Wetland and industrial areas provide a moderate level of thermal comfort in cold regions.
[72]	literature review	–	Researchers should develop indexes to explain the spatial and temporal variations of UHI.
[76]	Montreal	Simulation	In the daytime, the tree cover could reduce the air temperature at the tree level (4 °C at 20 m above the ground), as well as higher level (2 °C at 60 m above the ground).
[77]	Pavia	Field Measurements and Field survey	There is a strong correlation between indicators of thermal and acoustic comfort in indoor spaces.
[44]	Harbin	Field measurements and survey	In the summer season, traffic noise only causes a slight increase in thermal sensation while higher traffic noise led to lower thermal comfort in the three seasons.
[78]	Madrid	Field measurements and Field survey	Older people may be at risk due to thermal stress and high noise levels in cities.
[14]	Hong Kong	Field measurements and Field survey	People feel more comfortable if they are satisfied with aesthetics and acoustics.
[79]	China	Field measurements and Field survey	The acoustic and thermal environment and thermal environment are both important factors that affect subjective comfort.

comfort based on sound level measurements and questionnaire surveys has been investigated. Furthermore, the monitoring and survey in order to analyze the outdoor thermal comfort conditions were conducted simultaneously. Subsequently, possible correlations between the acoustic and thermal comfort parameters assessed were identified. Eventually, acoustic comfort differences based on thermal comfort parameters were analyzed in a park-based scale and index.

2. Methodology and data collection

This section describes the study area and how the thermal comfort and soundscape data of the EL Goli Park were collected and then analyzed.

2.1. Study area

This study was carried out in Tabriz, Iran (38.0792° N, 46.2887° E) at an altitude of 1351 m [80]. The existence of the Guzni and Bababaghi mountains in the northeast and Sahand mountains range in the south creates considerable differences in the climate in the north and west of the city. According to the Köppen classification, Tabriz has a cold and dry climate [81] with a maximum temperature of 26.60 °C, and a minimum temperature of −1.77, an average annual temperature of 12.9 °C (Fig. 1).

The selected location (El-Goli Park) with an approximate area of 70 km^2 hectares is located in the southeast region of Tabriz, Iran, which is surrounded by main highways in the south and west, and residential areas in the east and north. El-Goli Park is made up of a vast garden, a 40,000 m^2 artificial lake, an octagonal mansion in the center of the lake, and recreational spaces such as Luna Park and cafes that have been considered as an excellent example of a topographic design. This park was listed in the national heritage of Iran in 2009. The park includes a great diversity of natural communities, especially plant communities. The sounds of plants are one of the main features of the park, which can be found in different plant forms, leading to different biological soundscapes. To determine the selected points for the evaluation of outdoor acoustic and thermal comfort, first the climatic, cultural, and social characteristics of it were considered. Then, based on the type of plant enclosure, the selected spaces were classified into open spaces (OS), semi-open spaces (SS), closed spaces (CS) and vertical spaces (VS) (L1to L20). L1, L2, and L17 were located at the entrance of the park, where there were fewer plants and sound elements, but greater traffic. Similarly, L3 and L12, which are located inside the park, have a lower plant density. The remaining 15 selected points were surrounded by diverse plant and landscape characteristics (Table 2 and Fig. 1).

2.2. Objective assessment

2.2.1. Acoustic comfort data

Sound levels (dBA) were measured to analyze the quality of the acoustic environment at each observation point [28]. To achieve this, the A-weighting sound pressure level A (SPL) LAeq was recorded [82], with a sound level meter (see Appendix 1, Table 8) for a duration of 5 min. The meter microphone was positioned 1.2 m above ground level to decrease the impact of acoustic reflection [83]. Acoustic data collection was conducted for one week in June, July, and August 2021 during three specified time periods: 8–10 a.m., 1–3 p.m., and 6–8 p.m. The selected days had good weather and were without rain and wind (wind speed <10 m/s). Finally, to generate the representing LAeq, the mean value was calculated at all 20 locations.

2.2.2. Thermal comfort data

Microclimate data monitoring was conducted simultaneously in the time intervals mentioned in the previous section. To analyze and determine the range of the thermal comfort index in EL Goli park in Tabriz, PET was considered as a thermal comfort index. In this regard, given the ASHRAE standard, air temperature (Ta), relative humidity (Rh), wind speed (Ws), and mean radiant temperature (Mrt) have been analyzed. To calculate MRT, it was necessary to consider global temperature (TGLOBE) Eq. (1) where ε is global emissivity and D is global diameter [18,36,84] and was performed by the kestrel 5400 (standard ST-174B) while for calculating other parameters have been used a data logger at a height of 1.7 m has been used. Appendix 1, Table 8 illustrates the features of the equipment used.



Fig. 1. (a) Location of Tabriz city in Iran map of Koppen climate classification (Peel et al., 2007); (b) El-Goli Park in Tabriz city; (c) 20 selected points (plant spaces) within El-Goli Park; (d) diagram of plant space classification [58].

Table 2
Main characteristics of each selected point (receptors).

Selected points	Location	Space categories	Plant community
L1	Health road	Vertical space	Platanus, Populus nigra, Platanus acerifolia
L2	Pine forest	Vertical space	Pinus, Pine,
L3	The Green Hill	Vertical space	Populus, Platanus, Populus nigra,
L4	Greenwood	Vertical space	Populus, Acer, Platanus, Ailanthus
L5	Robinin forest	Closed space	Elaeagnus angustifolia, Acer, Robinia
L6	Acer forest	Closed space	Acer, Robinia, Olea europaea, Ailanthus, Elaeagnus angustifolia
L7	Arbor place	Closed space	Acer, Pinus, Carpinus betulus, Ailanthus, Morus alba
L8	El Goli artificial waterfall	Closed space	Elaeagnus angustifolia, Platanus, Pinus, Populus alba
L9	The Green Lake	Closed space	Salix babylonica, Acer, Pinus
L10	The Thousand Stairs	Semi-open space	Populus alba, Morus alba, Olea europaea
L11	Playing space	Semi-open space	Robinia, Acer, Salix babylonica
L12	Picnic space	Semi-open space	Pinus, Elaeagnus angustifolia
L13	playground	Semi-open space	Robinia, Morus alba, Fraxinus excelsior, Olea europaea
L14	Rest square	Semi-open space	Pinus, Berberis vulgaris, Acer, Salix babylonica
L15	Arbor place	Semi-open space	Pinus, Berberis vulgaris, Robinia
L16	Main Entrance	Open space	Buxus, Populus alba, Helichrysum bractatum
L17	Luna Park Entrance	Open space	Helichrysum bractatum, Ailanthus, Pinus
L18	El Goli Mansion	Open space	Pinus, Robinia, Helichrysum bractatum, Berberis vulgaris
L19	Sports space	Open space	Buxus, Berberis vulgaris, Morus alba, Oleaeuropaea
L20	Entrance	Open space	Buxus, Helichrysum bractatum

$$T_{MRT} = \left[(T_{GLOBE} + 273.15)^4 + \frac{1.1 * 10^8 * Ws^{0.6}}{\varepsilon * D^{0.4}} * (T_{GLOBE} - T_A) \right]^{0.25} - 273.15 \quad (1)$$

Furthermore, this index is not complete with climatic data and requires the activity of humans and the level of clothing. In this regard, the heat transfer resistance of the clothing and the type of activity of each participant were considered before calculating the PET using the RayMan software. The RayMan model, which was developed by Ref. [85], is used to calculate the physiological equivalent temperature (PET) and is very suitable for complex urban structures such as parks and urban plazas.

2.3. Subjective assessment (questionnaire)

Questionnaire surveys have been conducted simultaneously with the monitoring step, and it consists of 3 parts. The first part of the questionnaire is allocated to demographic data (that is, age and sex). In the second part, and given that prior research [25,86–88]; X. [24,29,66], three subsections were defined for acoustic comfort responses. Firstly, to examine the effects of perceived sound types on acoustic comfort, respondents were asked to rank three types of perceived sounds, using a ten-point Likert scale, in which 0 was 'unable to hear" and 10 was "clearly hear". Then, their answers were grouped according to the type of sound: natural, anthropogenic, or mechanical. Second, the participants were asked whether they perceived a subjectively dominant sound in the soundscape and if so, which. Then, four indicators including soundscape quality, the perceived loudness, landscape quality, and the overall impression were determined for participants to score on a scale of 1–5 (1 indicating the lowest quality and 5 indicating the highest). Eventually, the last section of the questionnaire was related to the visitors' position, type of activity, type of clothes as well as thermal sensation and thermal comfort. In this section, unlike many previous studies (M. [89]; M. A [90–92]. that used seven-category thermal sensation scales, due to the better adaptation to the wide range of outdoor thermal conditions [93] and more potential for the development of thermal comfort indices such as PET thermal sensation patterns [94,95] two extreme votes were added. Therefore, this scale ranges from −4 to +4 (very cold (−4), cold (−3), cool (−2), slightly cool (−1), neutral (0), slightly warm (+1), warm (+2), hot (+3) and very hot (+4). An example of the questionnaire used in the park is shown in Appendix 2, Table 9.

2.4. Characteristics of the respondents

Participants were randomly selected in the area, and a total of 300 valid questionnaires were collected. All participants were informed about the voluntary nature of the study before taking part. Considering that the differences in the basic physical environment might be caused by the time, the survey at the 20 positions was carried out from 9:00 a.m. to 21:00 a.m. Most visitors were sitting (45.7%) and the rest were walking (20.7%) or standing (33.6%). Moreover, respondents who were under the shading trees environment accounted for 43.26% while the rest of them were not in the shade 20 min before the survey. These data provide a comprehensive background of the respondents to ensure a random sampling of participants and allow potential adjustments to these factors in the statistical analyses. The age group that was repeated the most was 21–35 years (38.57%), followed by 36–50 years (28.7%). More men completed the survey than women (57% men, 43% women), and their average height and weight were 1.72 cm and 71 kg, respectively, while the average height and weight of the women who participated in this survey were 1.52 cm and 57 kg, respectively. A summary of personal information is observed in Table 3. These details give a rich portrait of the respondents, allowing for possible adjustments to be applied in statistical analyses to account for any imbalances in the distribution of respondents' characteristics.

2.5. Data analysis

SPSS 26.0 was used to establish a database of all subjective and objective results. Descriptive statistics were used to provide a summary of thermal and acoustic comfort evaluations. Kruskal-Wallis test was used to test if there were significant differences in thermal and acoustic comfort dimensions between four space categories. Pearson's correlation was conducted to test the correlations between the dimensions of acoustic and thermal comfort after being checked the normality of the data by the Kolmogorov-Smirnov (KS) test. Weighted linear regression models of the correlated indicators were developed. Furthermore, a one-way analysis of variance (ANOVA) test was conducted to find the relationships between thermal and acoustic comfort dimensions. Finally, simple linear regression was used to predict acoustic comfort based on

Table 3
Characteristics of the respondents.

Season	Gender	NO. of Persons	Age group (in %)		Average Height(m)	Average Weight (kg)
Summer	Male	171	Under 20	18.46%	1.72c.m	71 kg
	Female	129	21–35	38.57%	1.52 c.m	57 kg
			36–50	28.37%		
			51–60	10%		
			upper 60	4.6%		

thermal comfort indices on a spatial scale.

3. Results

3.1. Analysis of acoustic comfort and reported indicators

In terms of sound pressure level (SL), Fig. 2 shows the difference of LAeq among the selected places during summer. The existing differences are rooted in plant communities, user activities, and spatial conditions. **VS** with dense plants recorded the lowest sound levels. On the contrary, Location 1 reported a high LAeq (between 64 and 67 dB) due to the large gathering of birds, especially crows, as well as exercise activities. **CS** were the second quietest places and L8 reported remarkable low LAeq values with values between 55 and 58 dB, as it was adjacent to the residential parts and the protected forest covering the external noise. Meanwhile, **SS** showed a slight increase in sound levels. The high LAeq of place 11 reflected the impact of human activities, a place where morning sports, children's play, and cafe meetings occur. Finally, **OS** recorded the highest LAeqs (L16 and 20, with LAeq values between 66 and 72 dB), which was mainly due to exposure to external noise factors, including traffic and Luna Park equipment.

In terms of the identified sound types, Fig. 3 illustrates the frequency of the identified sound sources at each selected location. In **VS,** natural sounds were perceived the most and heard clearly, such as the swooshing sounds of dense plant species with larger leaves, such as plane trees. In addition, participants highly mentioned the sound of birds and water forms (burbling water, channels, and streams). In **CS,** both anthropogenic and natural sound were dominantly perceived. Nature sounds including rustling of trees with small leaves, such as willow (L19) and pine (L7 and L9), followed by bird singing, frogs, and cicadas was heard. Visitors also produce various types of human sounds such as talking, yelling, and footsteps. Regarding **SS,** human sounds were dominant and both natural and mechanical sound were secondly identified. The dominance of human sounds was due to various public activities including talking, playing, fitness activities, and cycling. Also, a weak swooshing can be heard where the plant community owned different leaf types, such as Robinia and Morus alba pendula (L13, L14 and L15). Bird singing was present at a lower level due to the lower density of plants. Lastly, **OS** owned the most frequent mechanical sounds, which were located near the Luna Park and city routs. Natural sounds were heard weakly, which were negligible at locations 19 and 20.

The Kruskal-Wallis test was performed to assess whether there were significant differences in acoustic comfort dimensions between the specified space types. Based on the results, as shown in Table 4, there were significant differences among the four indicators, including soundscape quality (SQ), overall impression (OI), percentage of reported natural dominant sound (%RNDS) and reported loudness (RL) ($p < .05$). However, there were no significant differences between LQ ($p > .05$). This finding could be explained by two factors: It is crucial to properly plan natural elements in line with other landscape aspects, such as architectural elements, because the sheer presence of natural elements, such as plants, cannot adequately boost the visual quality of the environment. Additionally, the landscape's vastness (the depth of field) is occasionally thought of as an environmental design potential that can be constrained by the presence of a plant community. For example, location 3 in the category of open space was said to have a high-quality environment because the El Goli Park mansion was located in the middle of the main pool and had beautiful views of the surroundings. Fig. 4 shows the mean values of reported indicators. It can be seen that SQ, OI and % RNDS showed a similar trend.

Regarding SQ, Table 4 shows that there were significant differences between VS and SS and OS ($p < .05$), between CS and OS ($p < .05$), and between SS and OS ($p < .05$), respectively. VS ($M = 4.07$) had the highest SQ value, while OS had the lowest values ($M = 2.26$). According to Fig. 3, it can be seen that SQ had a significant difference between VS with a rich natural community where natural sounds were perceived frequently and SS with fewer natural elements where human sounds were frequently perceived ($p < .05$), as well as OS with a weak existence of plants where mechanical sounds were predominantly reported ($p < .05$). However, no SQ difference was observed between CS (with dominant natural and human sounds) and VS (with dominant natural sounds), also between CS and SS (with dominant human sounds) ($p > .05$).

The perceived quality of OI might also be influenced by the composition of the plant spaces. As shown in Table 4, the OI was significantly different between VS and SS and OS ($p < .05$) and between CS with SS and OS ($p < .05$), while there was no difference between the others. According to Fig. 4, the OI of VS as well as CS with higher levels of the plant community had a significant difference with SS which has fewer natural elements and more human activities ($p < .05$), and, with OS which has the least plant cover and various mechanical sounds ($p < .05$). VS ($M = 4.11$) had the highest OI value, while OS had the lowest values ($M = 2.76$).

Regarding% RNDS, only subjectively natural dominant sounds were taken into consideration for the following analyses due to the results of numerous prior studies about the pleasantness of natural sounds and how it helps to create a nice soundscape. The results reported similar variations as the OI, which was significantly different between VS and SS and OS ($p < .05$) and between CS and SS and OS ($p < .05$) (Table 4). Higher percentages of natural dominant sounds were reported for VS and CS (identified by 52% and 41% of participants, respectively), which was rooted in dense plant cover, water forms and being located in the inner parts of the park. On the other hand, in SS due to the predominance of man-made sounds and in OS due to the masking effect of mechanical sounds, a lower% of RNDS was reported (29% and 25% of participants).

In terms of RL, there were differences between VS and OS ($p < .05$),

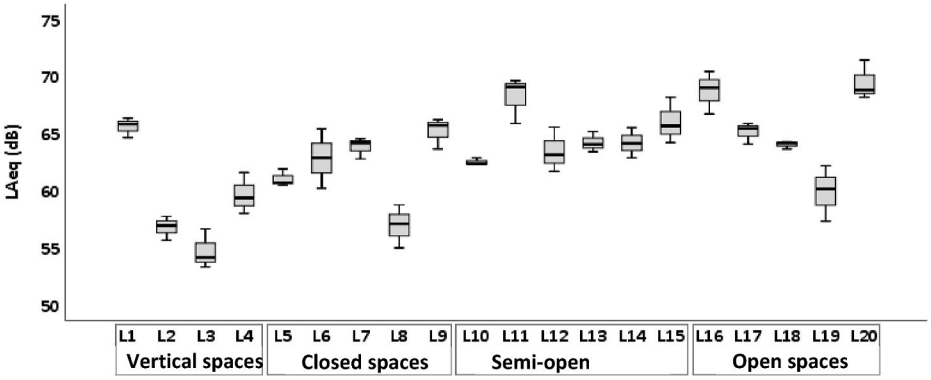


Fig. 2. Box plots representing the average measured values of LAeq within the space categories.

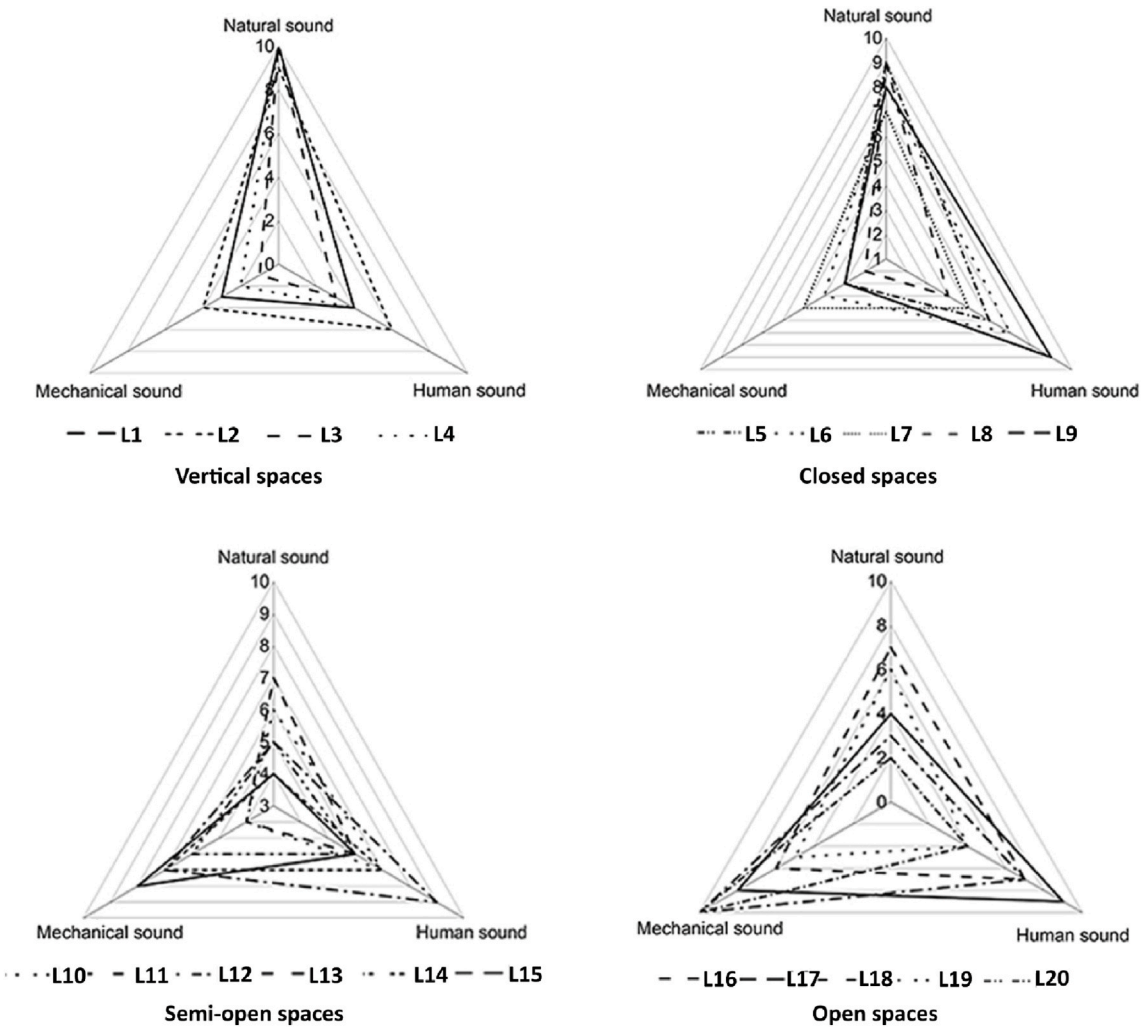


Fig. 3. The combination of perceived sounds in space categories.

Table 4
Differences in reported indicators among space categories, where ■, significant difference in SQ (soundscape quality). ●, significant difference in OI (overall picture). ◆, significant difference in RL (reported height). ▲, significant difference in RNDS % (reported natural dominant voice percentage).

Space types	VS	CS	SS	OS
VS	_		■*	■*
			●*	●*
			▲*	◆*
				▲*
CS		_	●*	■*
			◆*	●*
			▲*	◆*
				▲*
SS			_	■*
OS				_

VS = vertical spaces; CS = closed spaces; SS = semi-open spaces; OS = Open spaces. *p < .05.

and between CS with SS and OS ($p < .05$), while there were none among the others. There were lower levels of RL for VS and CS (M = 2.0 and 2.2 respectively), which could have rooted in the presence of multiple pleasant sounds in the environment. Meanwhile, there were higher levels of RL for SS and OS (M = 3.5 and 3.8 respectively), which could be due to the considerable presence of annoying sounds.

3.2. Analysis of outdoor thermal comfort

Regarding Ta, the evaluation of the results in El-Goli Park showed that the difference in Ta in different parts (OS, SS, CS and VS) is less than other indicators, so that the maximum difference in CS and SS is 0.41° Celsius. which is due to the difference of the plant community in the two spaces. However, Rh shows another trend from the Ta. CS and VS of the park due to high vaporization have the highest measure of Rh than the rest of the other parts. The shade of trees (both the tree canopy and height of the tree) is considered an influential indicator in the introduction of MRT which has been recorded due to being in the depth of the ground or the lack of suitable plants in some parts up to 57.3 °C. In this regard, VS experienced better conditions than other parts due to the placement of suitable crowns and tall trees, and the amount of MRT has decreased significantly. Although the average MRT difference between VS and OS has reached 5.52 °C due to lack of proper coverage. Evaluations made in connection with the impact of Ws on each part, it can be stated that, in the OS and VS, due to the lack of sufficient plant covers and the presence of tall trees with suitable coverage, respectively, led to

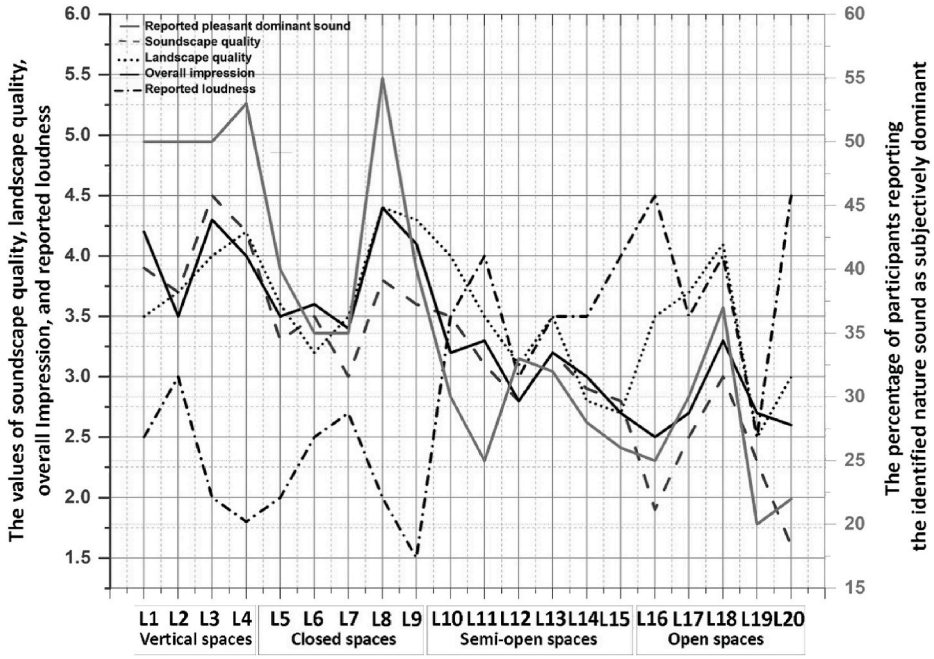


Fig. 4. The mean values of soundscape quality (SQ), landscape quality (LQ), overall impression (OI), reported loudness (RL) and the percentage of natural dominant sound (%RNDS).

creating more airflow than other parts.

In terms of PET, the most appropriate thermal comfort was observed in VS, where the average PET was 1.75 °C lower than the OS situation. The lack of adequate coverage in this part caused the most severe discomfort, and PET was recorded at 21.95 °C (Fig. 5). Separate analysis of sunny and night hours suggested that the maximum reduction in average daytime PET was equal to 3.38 °C reported when receptors were placed in VS. In addition, OS have the most intense thermal stress during sunny hours. However, the minimum PET was recorded in this scenario altering the thermal comfort condition to higher degrees of coolness. Fig. 5 demonstrated the mean changes in Ta, Rh, MRT, Ws, and PET at Z = 1.7 m entire measurement time.

Kruskal-Wallis test was conducted to identify significant differences in thermal comfort indicators (TSV and TCV) among the four space categories. According to the results, as shown in Table 5, there were significant differences in TSV and TCV ($p < .05$). The mean values of thermal sensation vote (TSV), thermal comfort vote (TCV), and psychological equivalent temperature (PET) are shown in Fig. 6. It can be seen that TSV and PET indicated similar variations.

TSV has also been influenced by the composition of plant spaces like PET. As demonstrated in Table 5, TSV was remarkably different between VS and SS and OS ($p < .05$) and between CS and OS ($p < .05$), while there was no difference between the other indicators. Our output has shown that TSV of VS and CS with higher levels of plant cover in comparison with SS and OS with fewer natural elements and more human activities had significant differences (Fig. 6). In other words, VS have recorded the most votes in neutral (45%) and slightly warm (30%). while warm (0.08%), slightly cool (0.08%), cool (0.05%) and cold (0.03%) followed. On the other hand, slightly warm was the most common vote in CS. Completely opposite conditions have been recorded in SS. The highest and lowest values of votes are assigned to warm (0.63%) and slightly warm (0.37%), respectively, in the SS. Finally, the most common vote in OS was hot (37%), followed by warm (34.6%). In this regard, while the overview was carried out in the hot season and there were noticeably more 'warm', 'hot' and 'very hot' votes, 'slightly warm' conditions were more common than 'warm', 'hot' and 'very hot' conditions because most of the participants were in shaded areas.

Regarding TCV, a significant difference has been observed between VS and SS and OS, and between CS and SS and OS, while there was no significant difference between other indicators (Table 5). According to Fig. 5, the highest and lowest level of TCV has been recorded in VS and OS respectively and CS and SS are in the next categories. During this project, about 2% of the participants in VS were very comfortable, then 4% were comfortable and 6% were very slightly comfortable with the thermal conditions. Although most of the participants in this area were slightly uncomfortable (44%) and neutral (36%) respectively. On the other hand, all participants felt slightly uncomfortable in CS. This situation has also continued in SS, with the difference that the number of people who feel uncomfortable (64%) is more than slightly uncomfortable (36%). Based on surveys of participants, conditions in OS have worsened, and most of the users feel very uncomfortable and uncomfortable with the thermal conditions, respectively.

To determine the outdoor thermal comfort range in Tabriz, PET [96] was used because of its frequent use as a thermal comfort index [62,67]. In this research, PET is calculated by RayMan [85] using measured Ta, RH, and WS and MRT. The clothing and activity are also taken into account in the PET calculation. To better identify the relationship between thermal comfort and thermal sensation vote (TSV), the mean thermal sensation votes (MTSV) were calculated for each 1 °C PET interval as in previous studies [62,97]. The correlation between PET and MTSV in this study is illustrated in Eq 3. Neutral PET in this climate is determined by substituting MTSV = 0 in Eq. 3. As a result, the neutral PET in this study is 25.85 °C. The acceptable thermal comfort range is between −0.5 and + 0.5 [98], so by replacing MTSV = ±0.5 in equation. 2, the outdoor thermal comfort in Tabriz ranges from 25.27 °C to 26.42 °C PET.

$$MTSV = 0.791PET\text{-}22.493 \quad (2)$$

3.3. Relationship between acoustic comfort and thermal comfort

An examination of the relationship between the reported thermal sensation vote (TSV) and the dimensions of acoustic comfort was done in an effort to confirm the hypothesis of the investigation. Furthermore, the relationship between acoustic comfort dimensions and thermal comfort

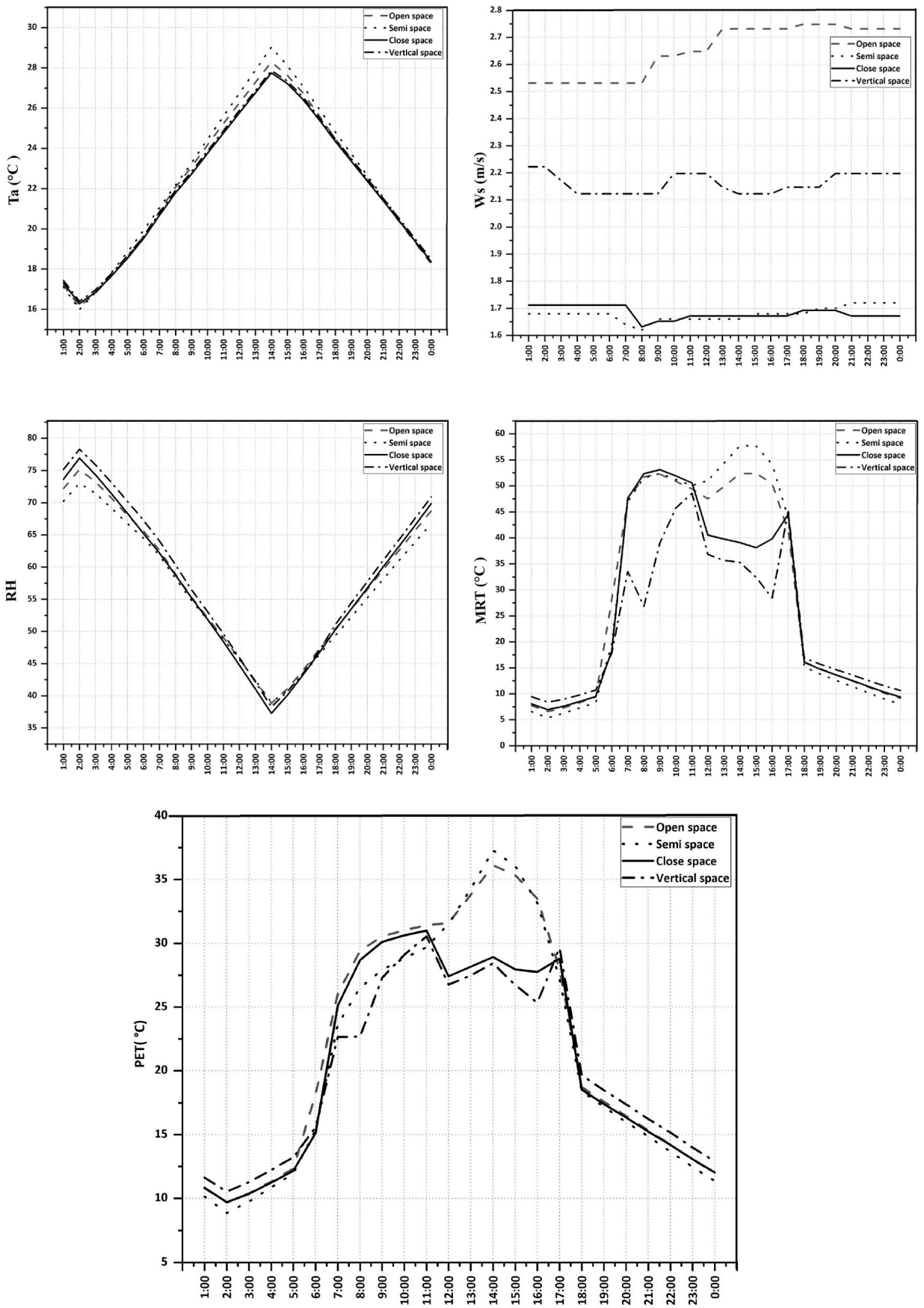


Fig. 5. The mean values of Ta, RH, Ws, MRT and PET.

Table 5
Difference in outdoor thermal comfort among space categories, where ■, with a significant difference in TVS (thermal sensation vote); ●, with a significant difference in TCV (thermal comfort vote).

Space types	VS	CS	SS	OS
VS	–		■* ●*	■* ●*
CS		–	●*	■* ●*
SS			–	
OS				–

VS = vertical spaces; CS = closed spaces; SS = semi-open spaces; OS = Open spaces. *p < .05.

vote (TCV) and physiological equivalent temperature (PET °C) was also studied. Table 6 displays the Pearson's correlation coefficients of the acoustic and thermal comfort metrics. The Kolmogorov-Smirnov test was used to determine the normality before Pearson's correlation analysis was performed. Of the six measured acoustic comfort indicators, four were found to be statistically significant with 'TSV', 'TCV' and 'PET' (p < .1). The results are depicted in Fig. 7 with the corresponding R2 values.

1. SQ (r = −0.875, p < .01), OI (r = −0.818, p < .01) and %RPDS (r = −0.795, p < .01) negatively, and RL (r = 0.604, p < .01) positively correlated with TSV. It was shown that more variance is explained when 'TSV' is used as a predictor for SQ ($R^2 = 0.76$), OI ($R^2 = 0.68$) and% RPDS ($R^2 = 0.63$), compared to RL ($R^2 = 0.36$). These findings support the hypothesis that visitors tended to report higher soundscape quality, higher overall impression, and higher percentage of natural dominant sound when they felt cooler in the environment. At the same time, these results reveal that a decrease in TSV can potentially reduce perceived loudness in the environment.
2. Furthermore, the Pearson product-moment correlation coefficient indicated a negative significant association between SQ (r = −0.739, p < .01), OI (r = −0.688, p < .01), %RPDS (r = −0.676, p < .01) and PET (°C) as well as positive significant association between RL and PET (°C) (r = 0.548, p < .01). The variation in PET (°C) among the four acoustic comfort indicators showed a similar trend, but generally there was reported a weaker effect was reported for PET (°C) in predicting the mean SQ ($R^2 = 0.54$), OI ($R^2 = 0.47$), %RPDS ($R^2 = 0.45$) and RL ($R^2 = 0.29$) compared to the other two indicators. Due

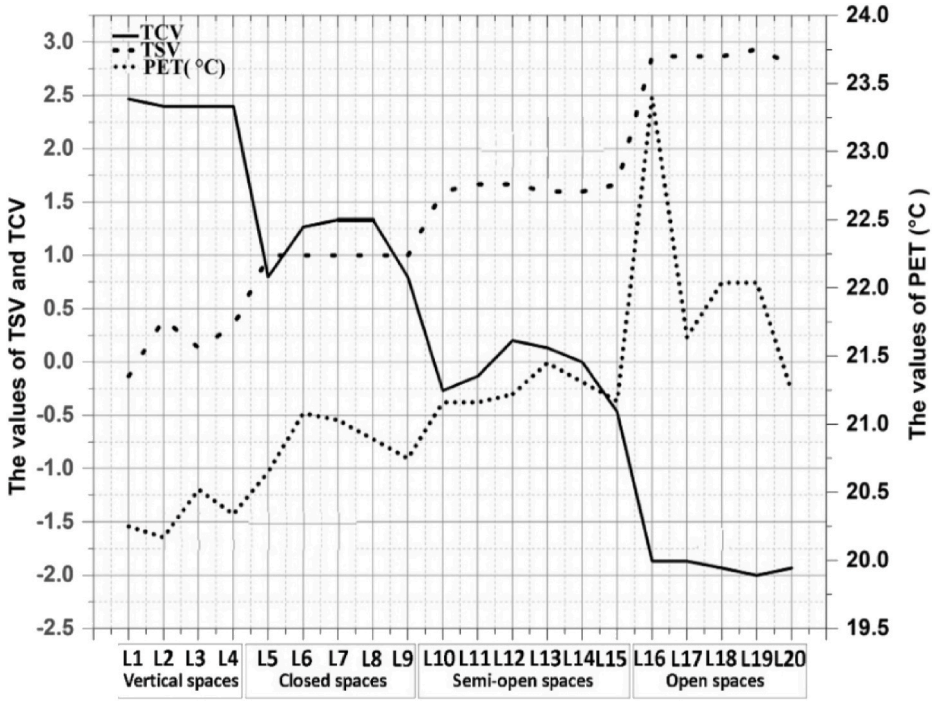


Fig. 6. Mean thermal comfort vote (TCV), thermal sensation vote (TSV), and physiological equivalent temperature (PET).

Table 6
Pearson's correlation (Pearson's r) between thermal comfort and acoustic comfort variables.

	SQ	OI	LQ	RL	%RPDS	SL	TSV	PET (°C)	TCV
SQ	1	0.895**	0.609**	−0.684**	0.846**	−0.413	−0.875**	−0.739**	0.869**
OI	–	1	0.688**	−0.728**	0.876**	−0.432	−0.818**	−0.688**	0.815**
LQ	–	–	1	−0.420	0.521**	−0.236	−0.372	−0.284	0.390
RL	–	–	–	1	−0.641**	0.721**	0.604**	0.548*	−0.619**
%RPDS	–	–	–	–	1	−0.423	−0.795**	−0.676**	0.826**
SL	–	–	–	–	–	1	0.333	0.298	−0.359
TSV	–	–	–	–	–	–	1	0.844**	−0.988**
PET (°C)	–	–	–	–	–	–	–	1	−0.824**
TCV	–	–	–	–	–	–	–	–	–

SQ = soundscape quality; OI = overall impression; LQ = landscape quality; RL = reported loudness; %RPDS = % reported pleasant dominant sound; SL = sound levels (dBA); TSV = thermal sensation vote; TCV = thermal comfort vote; PET (°C) = physiological equivalent temperature. ** The correlation is significant at the 0.01 level (2-tailed); * The correlation is significant at the 0.05 level (2-tailed).

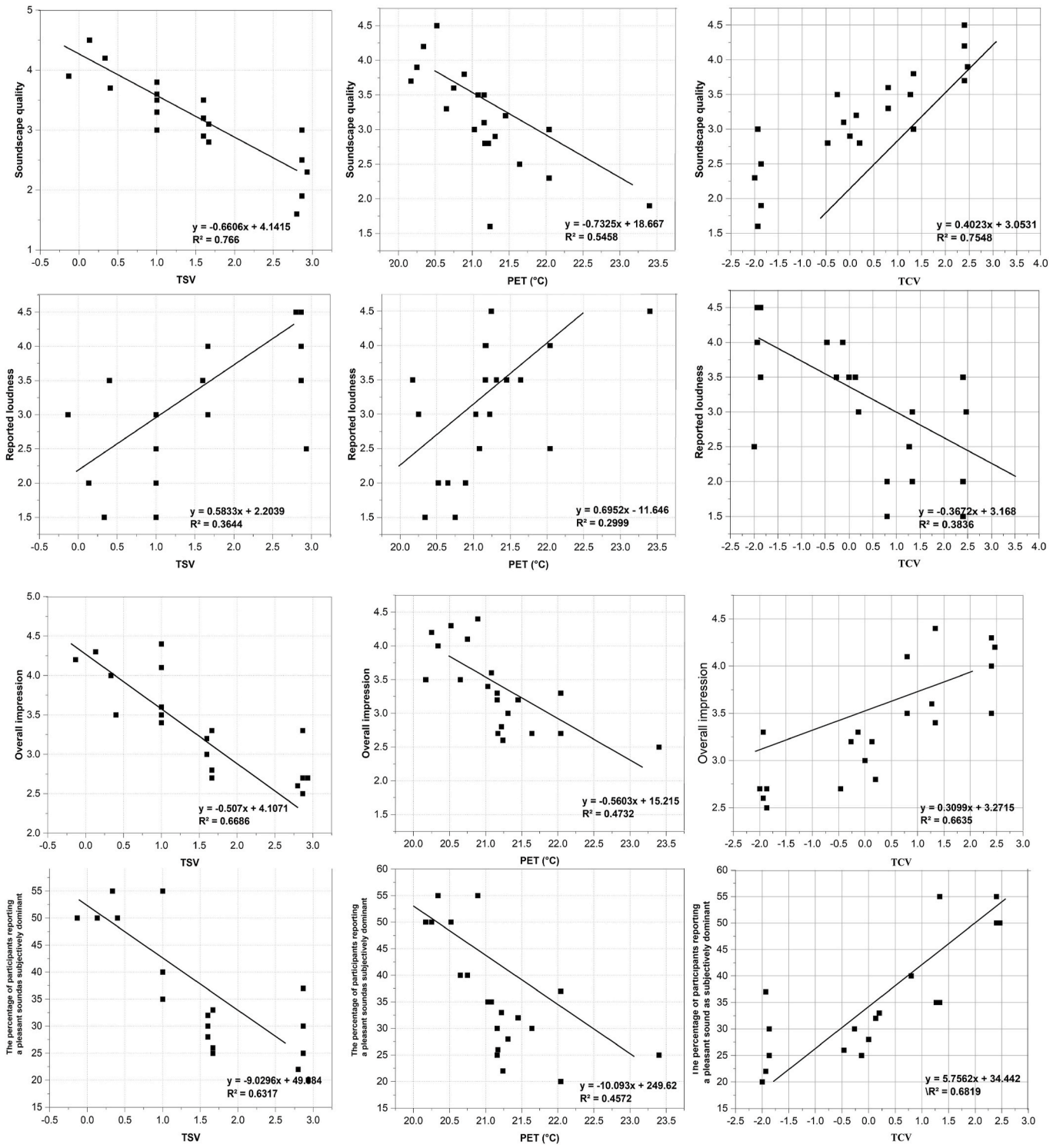


Fig. 7. Scatter plots between (a) mean TSV and the correlated indicators. Scatter plots between (b) mean PET and the correlated indicators. Scatterplots between (c) TCV and the correlated indicator.

to the same correlation variations of PET with TSV, similar interpretations stated for TSV are also true for PET (°C)).

3. In addition to TSV and PET parameters, strong positive correlations were detected between SQ ($r = 0.869$, $p < .01$), OI ($r = 0.815$, $p < .01$), %RPDS ($r = 0.826$, $p < .01$), and TCV, as well as negative correlations between RL ($r = -0.619$, $p < .01$) and TCV. Similarly, to the previous parts, there was a stronger relationship between SQ ($R^2 = 0.75$), OI ($R^2 = 0.65$), %RPDS ($R^2 = 0.68$) and TCV, compared to RL and TCV ($R^2 = 0.38$). The higher SQ, OI, and% RPDS, as the well as lower RL are perceived if the respondents present more thermal comfort votes.

Furthermore, it can be seen that the relationships between the two variables SL and LQ with each of the three variables of outdoor thermal comfort (TSV, PET (°C), and TCV) were almost insignificant ($p > .5$). This result suggests that the outdoor thermal comfort condition may not

affect the perceived landscape quality and the measured sound levels in the environment.

3.4. Acoustic comfort comparisons using an index-based and park-based approach

3.4.1. Index-based approach

To overcome the issue of spatial difference in the categorized positions, an index-based approach was considered. In this approach, the spatial scale covers all the 20 selected locations within El-Goli Park and the corresponding datasets. A one-way ANOVA test was conducted to examine whether the difference between average SQ, OI, % RPDS and RL across the groups of TSV/TCV groups (ie Warm, Neutral, Cool/Uncomfortable, Neutral, Comfortable) is significantly different. Levene's test was used to examine the homogeneity of variances, and it was discovered that it was violated in all instances since equal variances were not assumed.

- The analyzes showed that the average SQ is significantly different between the three TSV groups ($p < .005$) ($p < .005$). As shown in

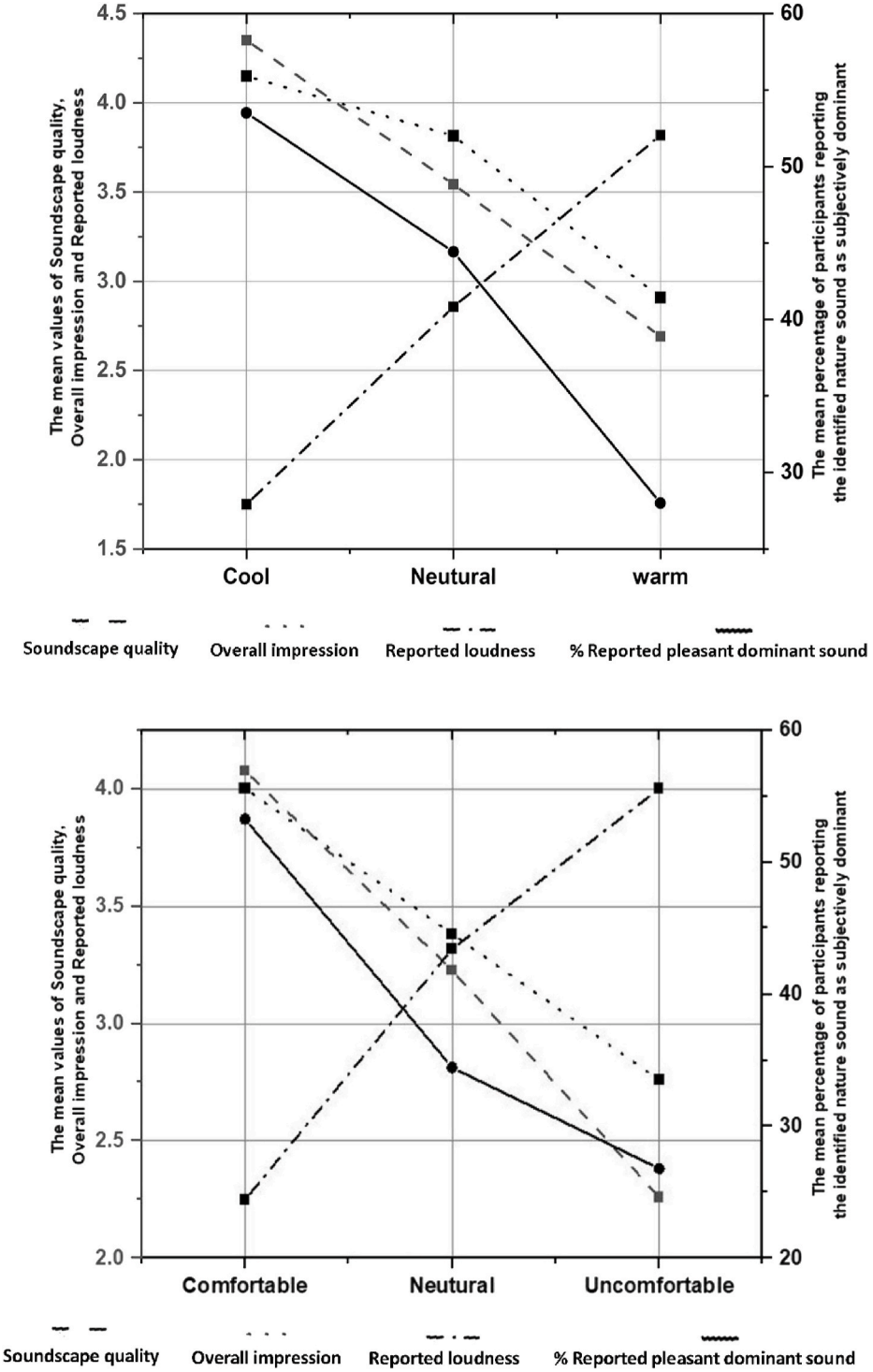


Fig. 8. Soundscape quality, overall impression, reported loudness, and percentage of reported pleasant dominant sound with different values of (a) TSV and (b) TCV.

Fig. 8a, the lowest mean SQ was in the 'Warm' group (2.6) compared to the 'Neutral' (3.5) and 'Cool' groups (4.4), respectively. These findings suggest that visitors who express warm, neutral, and cool TSVs perceive SQ in different ways to each other.

- Analysis of variance (ANOVA) demonstrated that the mean OI for the 'Warm' group differed significantly from the 'Neutral' and 'Cool' groups ($p < .005$), while there was no significant difference between the 'Neutral' and 'Cool' groups ($p > .05$). These findings show that among the TSV categories, feeling 'Warm' in the environment has the greatest impact on the evaluation of IAO, accounting for the lowest IAO value (2.8). In addition, visitors who feel cool and neutral in the environment perceive the quality of SQ in a similar way.
- The ANOVA test revealed similar findings to the OI and indicated that the mean %RPDS was significantly different between the 'Warm' group and the 'Neutral' and 'Cool' groups ($p < .05$). However, there were no differences between the 'Neutral' and 'Cool' groups ($p > .05$). This means that visitors who felt neutral in the environment presented a similar amount of %RPDS to those who felt cool. According to Fig. 8a, the highest calculated means of %RPDS was related to respondents who felt 'cool' (55%) and 'Neutral' (45%) with the thermal environment, respectively.
- There were significant differences in mean RL between three groups of TSV ($p < .005$). These findings indicate that visitors who rated the thermal environment as 'Warm' (3.8) perceived RL at higher levels than visitors who rated it as 'Neutral' (2.8) and 'Cool' (1.7), respectively.
  - ❖ On the other hand, the results, as in Fig. 8b, revealed that there are significant differences between the mean SQ of the three TCV groups ($p < .005$). The perceived quality of the SQ for the respondents who were 'uncomfortable' (2.2) with the thermal environment was low, compared to the groups 'neutral' (3.2) and 'Comfortable' (3.9) groups respectively, suggesting the potential effect of the quality of thermal comfort on sound environment perception.
  - ❖ The ANOVA test also showed that the perceived OI of the 'uncomfortable' groups was significantly different from the 'neutral' and 'Comfortable' groups ($p < .005$). However, it was not significant between the 'neutral' and 'Comfortable' group ($p > .05$), which implies that participants considering neutral TCV perceived OI in a similar way to those who were comfortable with the quality of thermal comfort. The mean measured OI for the 'Uncomfortable' and 'comfortable' group were 2.7 and 4.0 respectively (Fig. 8b).
  - ❖ It should also be noted that the analysis of variance performed for the %RPDS index was reported in the same way as the SQ index. As shown in Fig. 8b, the measured mean %RPDS for the 'Comfortable' and 'Uncomfortable' group were 54% and 26%, respectively.
  - ❖ According to the analyzes, there were significant differences in the mean RL of 'Comfortable' group with the 'neutral' and 'Uncomfortable' group ($p < .005$), while there was no significant difference between the 'neutral' and 'Uncomfortable' group ($p > .05$). This means that respondents who felt neutral or uncomfortable with the thermal environment perceived the loudness of the perceived sound to a similar extent. The lowest RL was reported for the 'Cool' group (2.2).

3.4.2. Park-based approach

VS, CS, SS and OS, as classified in Fig. 1, were analyzed with a simple linear regression test with regard to different plant communities. This process was conducted based on 3 predictors: a) TSV, b) TCV, and c) PET.

The effectiveness of the specified predictors is presented in Table 7 using the coefficient of determination (R2). From Table 7 it can be seen that for TCV and PET there was a very low coefficient of determination (R^2) ranging mostly between 0% and 14%. However, there were higher amounts of variance for TCV in predicting RL (between 0.13% and 52%). This means that the higher a visitor feels comfortable in the environment, the lower amount of subjective loudness is perceived, which is consistent with the respective findings in Fig. 8b. It is noteworthy that this effectiveness in open spaces allocated the highest R^2 (= 0.52). In the case of TSV, stronger variances have been reported, most of which are related to vertical spaces. Consequently, the effect of TSV in predicting SQ, %RNDS and RL is 45%, 55%, and 77%, respectively. However, very little variance was reported in terms of overall impression. It can be concluded that in VS with the highest amount of plant cover, the perceptual effect of thermal sensation plays an important role in the perceived quality of soundscape, the reported loudness, and the frequency of reported natural dominant sounds. This issue is less effective in closed, semi-open, and open spaces (Table 7).

4. Discussion

Outdoor acoustic comfort is an essential aspect of urban design that affects the use of outdoor spaces, the behavior of city dwellers, and the comfort felt in the environment. The increase in the growth of urbanization and the appearance of various types of noise pollution, including traffic and construction, weaken the acoustic comfort quality in the environment and cause users' dissatisfaction. However, different environmental factors, especially the thermal environment, can improve the acoustic conditions and the associated experiences in outdoor spaces. In the present study, the effect of outdoor thermal comfort on acoustic comfort and related indicators was investigated in four plant-space categories of an urban park (El-Goli, Tabriz).

Regarding the effects of thermal comfort on SQ, TSV and PET negatively and TCV positively are related to the perceived quality of soundscape. Our results indicated that, during summer, people in the cool and comfortable environments perceived a higher soundscape quality. It can be stated that by increasing the plant cover and adjusting the ambient temperature, the quality of the soundscape significantly will be promoted. In other words, the presence of natural elements in urban environments, especially trees, creates multiple positive soundscapes, such as wind blowing, rustling, birdsong, etc., which makes the environment more pleasant. These findings are similar to the results of previous studies [44,99,100]; M [34]. that low levels of thermal sensation (cooling of the environment) in the hot season and high levels of thermal comfort significantly increase the quality of the soundscape.

Regarding the effects on OI, similar to the SQ, TSV and PET negatively and TCV positively are related to the perceived quality of the

Table 7
The effect of each predictor (R2) on the respective acoustic comfort variables.

Space categories	Soundscape quality			Overall impression			Reported loudness			% Reported pleasant dominant sound		
	TSV	TCV	PET(°C)	TSV	TCV	PET(°C)	TSV	TCV	PET(°C)	TSV	TCV	PET(°C)
	R^2	R^2	R^2	R^2	R^2	R^2	R^2	R^2	R^2	R^2	R^2	R^2
VS	0.45	0.11	0.09	0.13	0.14	0.05	0.55	0.13	0.00	0.77	0.11	0.00
CS	0.31	0.01	0.03	0.05	0.00	0.04	0.19	0.43	0.05	0.09	0.02	0.07
SS	0.36	0.01	0.00	0.10	0.02	0.03	0.31	0.36	0.09	0.45	0.05	0.09
OS	0.20	0.00	0.00	0.01	0.06	0.03	0.34	0.52	0.03	0.06	0.12	0.00

overall environment. People in the cool and comfortable environments present a higher quality of overall impression. It can be stated that, better thermal comfort conditions by the presence of more natural elements have a significant effect on enhancing the OI quality and user's satisfaction as a result of creating a more comfortable, beautiful and pleasant space, which has been described by various authors [44,77,101].

Regarding the reported effects on %RDNS, TSV and PET negatively and TCV are positively related to the frequency of dominant natural sound. It showed that more %RDNS was reported in the groups with "cool" and "comfortable" thermal evaluations than in the groups with "warm" and "uncomfortable". It can be interpreted that the type of space configuration and the presence of extensive natural elements causes the perception of a higher percentage of natural sounds that prevail in the environment, which in the next stages strengthens user's perception and understanding of the soundscape and the environment [24,102]. In this regard [103], found that dominant natural sounds increase the pleasantness and eventfulness of perceived soundscapes.

Regarding the effects on RL, contrary to the previous sections, TSV, PET are positively and TCV negatively related to perceived loudness. It indicated that lower RL was reported in the groups with "cool" and "comfortable" thermal votes than in the groups with "warm" and "uncomfortable". Considering the presence of more greenery in cooler spaces, it is important to express the importance of the psychological effect of greenery in reducing the loudness of sound perceived by the users [104] Differently [44], found no relationship between thermal environment and perceived loudness. On the other hand, no correlation was found between thermal comfort and SPL (section 3.3). In this regard [78,105], suggested that there is no relationship between SPL and thermal perception indices, but there is relationship in case of loudness perceptions. It should be noted that all three thermal comfort indices showed slight influence on reported loudness compared to other dimensions of acoustic comfort.

Furthermore, as shown in Section 3.3, no correlation was found between thermal comfort and LQ. Several studies [14,106] stated that visual aesthetics can be effective in judging thermal comfort and users perceive higher comfort conditions in beautiful environments. However, according to the results of this study, in reverse mode, no relationship was found between thermal comfort and aesthetics. aesthetics. The reason can be rooted in the design method and composition of visual elements, including the presence of different visual elements, such as greenery, water forms, architectural aesthetics, and in some cases, the depth of the field of view. Similarly, the mean landscape quality in open and semi-open spaces is reported with a relatively slight difference compared to closed and vertical spaces.

The surveys were conducted in a limited time frame (summer 2021); hence, the findings may only be true under warm conditions. Furthermore, the role of other related variables such as the SVF index and noise indices may help the accuracy of the resulting analysis. Additionally, the focus of the current research was on the division of spaces based on plant communities, and the role of other effective factors, such as materials, water elements, etc., can be investigated in future studies to make the findings more accurate. Moreover, the factors that determine tree shading and show higher correlations with cooling effect and thermal comfort than with acoustic comfort can be analyzed in the future. Finally, more research is necessary at different locations considering their unique ecological and environmental characteristics and the social, economic, and cultural differences of their users.

5. Conclusions

Our study suggested that thermal comfort dimensions have significant effects on acoustic comfort and related indicators according to a questionnaire survey and thermal acoustic measurements. In this regard:

- Regarding TSV changes, the thermal sensation has a significant effect on SQ, %RNDS, OI, and RL. higher TSV results in a significant decrease in SQ, %RNDS, and OI. However, with the increase in TSV, RL increases significantly in all plant spaces. More precisely, the group in the 'Warm' thermal condition has a significantly lower SQ, %RNDS, and OI, as well as higher RL than the groups with 'Neutral' or 'Cool'. Furthermore, it was shown that TSV can be considered as a predictor for SQ, %RNDS, and RL in vertical spaces.
- With respect to changes in TCV, thermal comfort has a significant effect on SQ, %RNDS, OI, and RL. Higher TCV results in a significant increase in SQ, %RNDS, and OI. On the other hand, with the increase in TCV, RL decreases significantly. Specifically, the group under thermal condition has a significantly lower SQ,% RNDS and OI, than that of group with 'neutral' or 'Comfortable'. However, in terms of RL, the group in 'uncomfortable' thermal condition has a significantly higher RL only with the 'comfortable' group and RL is perceived in a similar way to the 'neutral' group. Furthermore, it was shown that TCV can be considered as a predictor of RL in plant spaces.
- With respect to PET changes, it can be said that PET has a significant effect on SQ, % RNDS, OI and RL. higher PET results in a significant decrease in SQ, %RNDS, and OI. On the other hand, with the increase in PET, RL increases significantly in all plant spaces. However, our analyzes showed that PET has lower effectiveness compared to TSV and TCV.

In addition, it should be noted that all three indicators of thermal comfort did not show any significant relationship with LQ and SPL. These findings are important for managing and improving outdoor acoustic comfort and soundscapes, since, for example, to improve a given soundscape, the outdoor thermal comfort condition should also be assessed too, and then, specific actions can be implemented to eliminate or reduce negative thermal comfort factors and promote positive factors. Therefore, more work is needed to identify the extensive and complex relationships of environmental factors to improve acoustic comfort in urban landscapes.

CRediT authorship contribution statement

Negar Mohammadzadeh: Writing – original draft, Visualization, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Alireza Karimi:** Writing – review & editing, Writing – original draft, Visualization, Investigation, Formal analysis, Data curation. **Robert D. Brown:** Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix 1

Table 8
Measured parameters and features of the instruments used in this research [62].

Instrument	Measured parameter	Accuracy	Range
Globe-thermometer	Tg,Ws	0.1 °C ° C at 0 °C DIN 43760 1/3	0.1
Weather datalogger	Ta	±0.9 °C from 40 °C to 60 °C ±0.5 °C from 5 °C to 40 °C ±1.1 °C from −20 °C to 5 °C ± 3%	−40 °C–70 °C
	Rh	±3%	0 to 90%
ST-885 sound level meter	LAeq	±1.4 dBA	30–130 dBA; 31.5 Hz-8 kHz.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.buildenv.2022.109884.

References

[1] R. Reitano, Water harvesting and water collection systems in Mediterranean area. The case of Malta, Procedia Eng. 21 (2011) 81–88.
[2] J. Adusei-Gyamfi, K.S. Boateng, A. Sulemana, J.N. Hogarh, Post COVID-19 recovery: challenges and opportunities for solid waste management in Africa, Environmental Challenges (2022), 100442.
[3] F.O. Ajibade, B. Adelodun, K.H. Lasisi, O.O. Fadare, T.F. Ajibade, N.A. Nwogwu, I.D. Sulaymon, A.Y. Ugya, H.C. Wang, A. Wang, Environmental pollution and their socioeconomic impacts, in: Microbe Mediated Remediation of Environmental Contaminants, Elsevier, 2021, pp. 321–354.
[4] S. Wang, D. Zhou, Architectural plasticity in response to population density in Abutilon theophrasti (Malvaceae), Ecol. Res. 37 (2) (2022) 228–239.
[5] L. Goines, L. Hagler, Noise pollution: a modem plague, South. Med. J. 100 (3) (2007) 287–294.
[6] A. Karimi, P. Mohammad, A. García-Martínez, D. Moreno-Rangel, D. Gachkar, S. Gachkar, New developments and future challenges in reducing and controlling heat island effect in urban areas, Environ. Dev. Sustain. (2022) 1–47.
[7] A. Gupta, A. Gupta, K. Jain, S. Gupta, Noise pollution and impact on children health, Indian J. Pediatr. 85 (4) (2018) 300–306.
[8] R. Thompson, R.B. Smith, Y.B. Karim, C. Shen, K. Drummond, C. Teng, M. B. Toledano, Noise pollution and human cognition: an updated systematic review and meta-analysis of recent evidence, Environ. Int. 158 (2022), 106905.
[9] B. Halder, A. Karimi, P. Mohammad, J. Bandyopadhyay, R.D. Brown, Z. M. Yaseen, Investigating the relationship between land alteration and the urban heat island of Seville city using multi-temporal Landsat data, Theor. Appl. Climatol. (2022) 1–23.
[10] L. Loures, R. Santos, T. Panagopoulos, Urban parks and sustainable city planning-The case of Portimão, Portugal, Population 15 (10) (2007) 171–180.
[11] M. Braubach, A. Egorov, P. Mudu, T. Wolf, C. Ward Thompson, M. Martuzzi, Effects_of_Urban_Green_Space_on_Environmental_Heal, 2017, pp. 51–64, https://doi.org/10.1007/978-3-319-56091-5.
[12] I.B. Campos, R. Fewster, T. Landers, A. Truskinger, M. Towsey, P. Roe, B. Lee, A. Gaskett, Acoustic region workflow for efficient comparison of soundscapes under different invasive mammals' management regimes, Ecol. Inf. (2022), 101554.
[13] Y. Geng, B. Hong, M. Du, T. Yuan, Y. Wang, Combined effects of visual-acoustic-thermal comfort in campus open spaces: a pilot study in China's cold region, Build. Environ. 209 (2022), 108658.
[14] K.K.-L. Lau, C.Y. Choi, The influence of perceived aesthetic and acoustic quality on outdoor thermal comfort in urban environment, Build. Environ. 206 (2021), 108333.
[15] J. Liu, Y. Xiong, Y. Wang, T. Luo, Soundscape effects on visiting experience in city park: a case study in Fuzhou, China, Urban For. Urban Green. 31 (January) (2018) 38–47, https://doi.org/10.1016/j.ufug.2018.01.022.
[16] N.K. Afshar, Z. Karimian, R. Doostan, M.H. Nokhandan, Influence of planting designs on winter thermal comfort in an urban park, J. Environ. Eng. Landsc. Manag. 26 (3) (2018) 232–240.
[17] M. Afsharzadeh, M. Khorasanizadeh, S. Norouzian-Maleki, A. Karimi, Identifying and prioritizing the design attributes to improve the use of besat park of tehran, Iran, Iran Univ. Sci. Technol. 31 (3) (2021) 1–24.
[18] P. Mohammad, S. Aghlmand, A. Fadaei, S. Gachkar, D. Gachkar, A. Karimi, Evaluating the role of the albedo of material and vegetation scenarios along the urban street canyon for improving pedestrian thermal comfort outdoors, Urban Clim. 40 (2021), 100993, https://doi.org/10.1016/j.uclim.2021.100993.
[19] X. Zhang, M. Ba, J. Kang, Q. Meng, Effect of soundscape dimensions on acoustic comfort in urban open public spaces, Appl. Acoust. 133 (November 2017) (2018) 73–81, https://doi.org/10.1016/j.apacoust.2017.11.024.
[20] B.M. Brooks, B. Schulte-Fortkamp, K.S. Voigt, Alex U. Case, Exploring our sonic environment through soundscape research & theory, Acoust. Today 10 (1) (2014) 30–40, https://doi.org/10.1121/1.4870174.
[21] F. Liu, J. Kang, A grounded theory approach to the subjective understanding of urban soundscape in Sheffield, Cities 50 (2016) 28–39.
[22] ISO 12913-1, Acoustics — Soundscape — Part 1: Definition and Conceptual Framework, 2014.
[23] Q. Meng, J. Kang, H. Jin, Field study on the influence of spatial and environmental characteristics on the evaluation of subjective loudness and acoustic comfort in underground shopping streets, Appl. Acoust. 74 (8) (2013) 1001–1009.
[24] G. Pérez-Martínez, A.J. Torija, D.P. Ruiz, Soundscape assessment of a monumental place: a methodology based on the perception of dominant sounds, Landsc. Urban Plann. 169 (December 2016) (2018) 12–21, https://doi.org/10.1016/j.landurbplan.2017.07.022.
[25] F. Aletta, J. Kang, Ö. Axelsson, Soundscape descriptors and a conceptual framework for developing predictive soundscape models, Landsc. Urban Plann. 149 (2016) 65–74, https://doi.org/10.1016/j.landurbplan.2016.02.001.
[26] F. Aletta, Ö. Axelsson, J. Kang, Dimensions underlying the perceived similarity of acoustic environments, Front. Psychol. 8 (2017) 1162.
[27] Ö. Axelsson, M.E. Nilsson, B. Hellström, P. Lundén, A field experiment on the impact of sounds from a jet-and-basin fountain on soundscape quality in an urban park, Landsc. Urban Plann. 123 (2014) 49–60.
[28] J. Liu, J. Kang, T. Luo, H. Behm, Landscape effects on soundscape experience in city parks, Sci. Total Environ. 454 (455) (2013) 474–481, https://doi.org/10.1016/j.scitotenv.2013.03.038.
[29] J. Liu, J. Kang, T. Luo, H. Behm, T. Coppack, Spatiotemporal variability of soundscapes in a multiple functional urban area, Landsc. Urban Plann. 115 (2013) 1–9.
[30] W. Yang, J. Kang, Acoustic comfort evaluation in urban open public spaces, Appl. Acoust. 66 (2005) 211–229, https://doi.org/10.1016/j.apacoust.2004.07.011.
[31] L. Yu, Soundscape Evaluation and ANN Modelling in Urban Open Spaces, University of Sheffield, 2009.
[32] J.Y. Hong, J.Y. Jeon, Exploring spatial relationships among soundscape variables in urban areas: a spatial statistical modelling approach, Landsc. Urban Plann. 157 (2017) 352–364.
[33] S. Marry, J. Defrance, Analysis of the perception and representation of sonic public spaces through on site survey, acoustic indicators and in-depth interviews, Appl. Acoust. 74 (2) (2013) 282–292.
[34] M. Zhang, J. Kang, Towards the evaluation, description, and creation of soundscapes in urban open spaces, Environ. Plann. Plann. Des. 34 (1) (2007) 68–86.
[35] M. Cetin, The changing of important factors in the landscape planning occur due to global climate change in temperature, Rain and climate types: a case study of Mersin City, Turkish J. Agric. Food Sci. Technol. 8 (12) (2020) 2695–2701.
[36] N. Narimani, A. Karimi, R.D. Brown, Effects of street orientation and tree species thermal comfort within urban canyons in a hot, dry climate, Ecol. Inf. (2022), 101671.
[37] Y.Y. Lee, M.F.M. Din, M. Ponraj, Z.Z. Noor, K. Iwao, S. Chelliapan, Overview of urban heat island (uhi) phenomenon towards human thermal comfort, Environ. Eng. Manag. J. (EEMJ) 16 (9) (2017).
[38] R.D. Brown, T.J. Gillespie, Estimating outdoor thermal comfort using a cylindrical radiation thermometer and an energy budget model, Int. J. Biometeorol. 30 (1) (1986) 43–52, https://doi.org/10.1007/BF02192058.
[39] N. Mazhar, R.D. Brown, N. Kenny, S. Lenzholzer, Thermal comfort of outdoor spaces in Lahore, Pakistan: lessons for bioclimatic urban design in the context of global climate change, Landsc. Urban Plann. 138 (2015) 110–117, https://doi.org/10.1016/j.landurbplan.2015.02.007.
[40] S.Y. Chan, C.K. Chau, T.M. Leung, On the study of thermal comfort and perceptions of environmental features in urban parks: a structural equation modeling approach, Build. Environ. 122 (2017) 171–183, https://doi.org/10.1016/j.buildenv.2017.06.014.
[41] T. Galindo, M.A. Hermida, Effects of thermophysiological and non-thermal factors on outdoor thermal perceptions: the Tomebamba Riverbanks case, Build. Environ. 138 (2018) 235–249.

[42] H. Guan, S. Hu, M. Lu, M. He, Z. Mao, G. Liu, People's subjective and physiological responses to the combined thermal-acoustic environments, Build. Environ. 172 (2020), 106709.
[43] K. Nagano, T. Horikoshi, New index of combined effect of temperature and noise on human comfort: summer experiments on hot ambient temperature and traffic noise, Arch. Complex Environ. Stud. 13 (3–4) (2001).
[44] Y. Jin, H. Jin, J. Kang, Combined effects of the thermal-acoustic environment on subjective evaluations in urban squares, Build. Environ. 168 (2020), 106517.
[45] Wonyoung Yang, H.J. Moon, Cross-modal effects of noise and thermal conditions on indoor environmental perception and speech recognition, Appl. Acoust. 141 (2018) 1–8.
[46] S. Louafi, S. Abdou, S. Reiter, Effect of vegetation cover on thermal and visual comfort of pedestrians in urban spaces in hot and dry climate, Nat. Technol. 17 (2017) 30B.
[47] A.M. Bustamante-Zapata, C.A. Zafra-Mejía, H.A. Rondón-Quintana, Influence of vegetation on outdoor thermal comfort in a high-altitude tropical megacity: climate change and variability scenarios, Buildings 12 (5) (2022) 520.
[48] A. Habibi, R. Tafvizi, R. Vakilinejad, M. Sheibani, The effect of vegetation and topography on thermal comfort in urban spaces, Geogr. Territ. Spatial Arrang. 12 (42) (2022).
[49] J. Zhang, F. Zhang, Z. Gou, J. Liu, Assessment of macroclimate and microclimate effects on outdoor thermal comfort via artificial neural network models, Urban Clim. 42 (2022), 101134.
[50] H.S. Jang, S.C. Lee, J.Y. Jeon, J. Kang, Evaluation of road traffic noise abatement by vegetation treatment in a 1: 10 urban scale model, J. Acoust. Soc. Am. 138 (6) (2015) 3884–3895.
[51] E. Margaritis, J. Kang, K. Filipan, D. Botteldooren, The influence of vegetation and surrounding traffic noise parameters on the sound environment of urban parks, Appl. Geogr. 94 (2018) 199–212.
[52] T. Van Renterghem, K. Attenborough, M. Maennel, J. Defrance, K. Horoshenkov, J. Kang, I. Bashir, S. Taherzadeh, B. Altreuther, A. Khan, Measured light vehicle noise reduction by hedges, Appl. Acoust. 78 (2014) 19–27.
[53] E. Margaritis, J. Kang, Relationship between urban green spaces and other features of urban morphology with traffic noise distribution, Urban For. Urban Green. 15 (2016) 174–185.
[54] G. Watts, A. Miah, R. Pheasant, Tranquillity and soundscapes in urban green spaces—predicted and actual assessments from a questionnaire survey, Environ. Plann. Plann. Des. 40 (1) (2013) 170–181.
[55] A.M. Dzhambov, D.D. Dimitrova, Green spaces and environmental noise perception, Urban For. Urban Green. 14 (4) (2015) 1000–1008.
[56] F. Yang, Z.Y. Bao, Z.J. Zhu, An assessment of psychological noise reduction by landscape plants, Int. J. Environ. Res. Publ. Health 8 (4) (2011) 1032–1048.
[57] G. Brambilla, V. Gallo, F. Asdrubali, F. D'Alessandro, The perceived quality of soundscape in three urban parks in Rome, J. Acoust. Soc. Am. 134 (1) (2013) 832–839, https://doi.org/10.1121/1.4807811.
[58] X. Song, X. Lv, D. Yu, Q. Wu, Spatial-temporal change analysis of plant soundscapes and their design methods, Urban For. Urban Green. 29 (2018) 96–105.
[59] M. Weber, The soundscape of nature areas: assessment and review of research approaches, Acoustics 2012 (2012).
[60] E. Gómez-Baggethun, D.N. Barton, Classifying and valuing ecosystem services for urban planning, Ecol. Econ. 86 (2013) 235–245.
[61] D. Armson, P. Stringer, A.R. Ennos, The effect of tree shade and grass on surface and globe temperatures in an urban area, Urban For. Urban Green. 11 (3) (2012) 245–255.
[62] N. Nasrollahi, Y. Namazi, M. Taleghani, The effect of urban shading and canyon geometry on outdoor thermal comfort in hot climates: a case study of Ahvaz, Iran, Sustain. Cities Soc. 65 (2021), 102638.
[63] S.N. Kandelan, M. Yeganeh, S. Peyman, K. Panchabikesan, U. Eicker, Environmental study on greenery planning scenarios to improve the air quality in urban canyons, Sustain. Cities Soc. (2022), 103993.
[64] H. Liu, J.Y. Lim, B.W.H. Thet, P.-Y. Lai, W.S. Koh, Evaluating the impact of tree morphologies and planting densities on outdoor thermal comfort in tropical residential precincts in Singapore, Build. Environ. (2022), 109268.
[65] J. Niu, J. Xiong, H. Qin, J. Hu, J. Deng, G. Han, J. Yan, Influence of thermal comfort of green spaces on physical activity: empirical study in an urban park in Chongqing, China, Build. Environ. 219 (2022), 109168.
[66] L. Zhang, Q. Zhan, Y. Lan, Effects of the tree distribution and species on outdoor environment conditions in a hot summer and cold winter zone: a case study in Wuhan residential quarters, Build. Environ. 130 (September 2017) (2018) 27–39, https://doi.org/10.1016/j.buildenv.2017.12.014.
[67] A. Karimi, P. Mohammad, Effect of outdoor thermal comfort condition on visit of tourists in historical urban plazas of Sevilla and Madrid, Environ. Sci. Pollut. Control Ser. (2022) 1–21.
[68] S. Oliveira, H. Andrade, T. Vaz, The cooling effect of green spaces as a contribution to the mitigation of urban heat: a case study in Lisbon, Build. Environ. 46 (11) (2011) 2186–2194.
[69] C. Romeo, M. Zinzi, Impact of a cool roof application on the energy and comfort performance in an existing non-residential building. A Sicilian case study, Energy Build. 67 (2013) 647–657.
[70] A. Synnefa, T. Karlessi, N. Gaitani, M. Santamouris, D.N. Assimakopoulos, C. Papakatsikas, Experimental testing of cool colored thin layer asphalt and estimation of its potential to improve the urban microclimate, Build. Environ. 46 (1) (2011) 38–44.
[71] S. Yilmaz, M.A. Irmak, A. Qaid, Assessing the effects of different urban landscapes and built environment patterns on thermal comfort and air pollution in Erzurum city, Turkey, Build. Environ. (2022), 109210.
[72] S.W. Kim, R.D. Brown, Pedestrians' behavior based on outdoor thermal comfort and micro-scale thermal environments, Austin, TX, Sci. Total Environ. 808 (2022), 152143.
[73] Y. Wang, U. Berardi, H. Akbari, Comparing the effects of urban heat island mitigation strategies for Toronto, Canada, Energy Build. 114 (2016) 2–19.
[74] M. Sheikh, A. Mitchell, Design strategies for perceived acoustic comfort in urban environments–A literature review, Proc. ACOUSTICS 7 (9) (2018).
[75] Q. Weng, M.K. Firozjaei, A. Sedighi, M. Kiavarz, S.K. Alavipanah, Statistical analysis of surface urban heat island intensity variations: a case study of babol city, Iran, GIScience Remote Sens. 56 (4) (2019) 576–604.
[76] Y. Wang, H. Akbari, Analysis of urban heat island phenomenon and mitigation solutions evaluation for Montreal, Sustain. Cities Soc. 26 (2016) 438–446.
[77] C. Buratti, E. Belloni, F. Merli, P. Ricciardi, A new index combining thermal, acoustic, and visual comfort of moderate environments in temperate climates, Build. Environ. 139 (2018) 27–37.
[78] M.T.B. Larriva, E. Higueras, Health risk for older adults in Madrid, by outdoor thermal and acoustic comfort, Urban Clim. 34 (2020), 100724.
[79] Y. Yin, D. Zhang, M. Zhen, W. Jing, W. Luo, W. Feng, Combined effects of the thermal-acoustic environment on subjective evaluations in outdoor public spaces, Sustain. Cities Soc. 77 (2022), 103522.
[80] G. Roshan, R. Oji, P. Fdez-Arróyabe, Future impact of cloudiness and wind changes on thermal comfort conditions in the city of Tabriz (Iran) in the frame of climate change, Sustain. Cities Soc. 77 (2022), 103575.
[81] S. Teshnehdel, H. Akbari, E. Di Giuseppe, R.D. Brown, Effect of tree cover and tree species on microclimate and pedestrian comfort in a residential district in Iran, Build. Environ. 178 (May) (2020), https://doi.org/10.1016/j.buildenv.2020.106899.
[82] J.Y. Jeon, J.Y. Hong, Classification of urban park soundscapes through perceptions of the acoustical environments, Landsc. Urban Plann. 141 (2015) 100–111.
[83] P. Zahorik, Assessing auditory distance perception using virtual acoustics, Acoust. Soc. Am. 111 (4) (2002) 1832–1846, https://doi.org/10.1121/1.1458027.
[84] S.H. Rizvi, H. Fatima, M.J. Iqbal, K. Alam, The effect of urbanization on the intensification of SUHIs: analysis by LULC on Karachi, J. Atmos. Sol. Terr. Phys. 207 (June) (2020), 105374, https://doi.org/10.1016/j.jastp.2020.105374.
[85] A. Matzarakis, F. Rutz, H. Mayer, Modelling radiation fluxes in simple and complex environments - application of the RayMan model, Int. J. Biometeorol. 51 (4) (2007) 323–334, https://doi.org/10.1007/s00484-006-0061-8.
[86] J. Kang, W. Yang, Soundscape and sound preferences in urban squares : a case study in sheffield soundscape and sound preferences in urban squares : a case study in sheffield, J. Urban Des. 10 (1) (2005) 61–80, https://doi.org/10.1080/13574800500062395.
[87] J. Liu, Y. Wang, C. Zimmer, J. Kang, T. Yu, Factors associated with soundscape experiences in urban green spaces: a case study in Rostock, Germany, Urban For. Urban Green. 37 (July 2017) (2019) 135–146, https://doi.org/10.1016/j.ufug.2017.11.003.
[88] J. Liu, L. Yang, Y. Xiong, Y. Yang, Effects of soundscape perception on visiting experience in a renovated historical block, Build. Environ. 165 (August) (2019), 106375, https://doi.org/10.1016/j.buildenv.2019.106375.
[89] M. Humphreys, F. Nicol, S. Roaf, Adaptive Thermal Comfort: Foundations and Analysis, Routledge, 2015.
[90] M.A. Humphreys, J.F. Nicol, I.A. Raja, Field studies of indoor thermal comfort and the progress of the adaptive approach, in: Advances in Building Energy Research, Routledge, 2012, pp. 55–88.
[91] E.L. Krüger, F.O. Minella, F. Rasia, Impact of urban geometry on thermal comfort and air quality from field measurements in Curitiba, Braziloutdoor, Build. Environ. 46 (3) (2011) 621–634.
[92] J. Nakano, S.-I. Tanabe, Thermal comfort and adaptation in semi-outdoor environments, Build. Eng. 110 (2004) 543.
[93] I. Knez, S. Thorsson, Thermal, emotional and perceptual evaluations of a park: cross-cultural and environmental attitude comparisons, Build. Environ. 43 (9) (2008) 1483–1490.
[94] N. Kántor, L. Égerházi, J. Unger, Subjective estimation of thermal environment in recreational urban spaces—part 1: investigations in Szeged, Hungary, Int. J. Biometeorol. 56 (6) (2012) 1075–1088.
[95] N. Kántor, A. Kovács, Á. Takács, Seasonal differences in the subjective assessment of outdoor thermal conditions and the impact of analysis techniques on the obtained results, Int. J. Biometeorol. 60 (11) (2016) 1615–1635.
[96] P. Höppe, The physiological equivalent temperature - a universal index for the biometeorological assessment of the thermal environment, Int. J. Biometeorol. 43 (2) (1999) 71–75, https://doi.org/10.1007/s004840050118.
[97] A. Karimi, H. Sanaieian, H. Farhadi, S. Norouzian-Maleki, Evaluation of the thermal indices and thermal comfort improvement by different vegetation species and materials in a medium-sized urban park, Energy Rep. 6 (2020) 1670–1684.
[98] R. Mora, R. Bean, Thermal comfort: designing for people, ASHRAE J. 60 (2) (2018) 40–46.
[99] Y. Hao, J. Kang, H. Wörtche, Assessment of the masking effects of birdsong on the road traffic noise environment, J. Acoust. Soc. Am. 140 (2) (2016) 978–987.
[100] L. Yu, J. Kang, Modeling subjective evaluation of soundscape quality in urban open spaces: an artificial neural network approach, J. Acoust. Soc. Am. 126 (3) (2009) 1163–1174.
[101] Q. Huang, X. Meng, X. Yang, L. Jin, X. Liu, W. Hu, The ecological city: considering outdoor thermal environment, Energy Proc. 104 (2016) 177–182.

[102] Y.G. Matsinos, A.D. Mazaris, K. Papadimitriou, A. Mniestris, Spatio-temporal variability in human and natural sounds in a rural landscape Spatio-temporal variability in human and natural sounds in a rural landscape, Landsc. Ecol. 23 (8) (2008) 945–959, https://doi.org/10.1007/s10980-008-9250-7.
[103] Ö. Axelsson, M.E. Nilsson, B. Berglund, A principal components model of soundscape perception a), J. Acoust. Soc. Am. 128 (5) (2010) 2846, https://doi.org/10.1121/1.3493436, 2846.
[104] H. Kuttruff, Acoustics: an Introduction, CRC Press, 2007.
[105] K. Nagano, T. Horikoshi, New comfort index during combined conditions of moderate low ambient temperature and traffic noise, Energy Build. 37 (3) (2005) 287–294.
[106] N. Pellerin, V. Candas, Combined effects of temperature and noise on human discomfort, Physiol. Behav. 78 (1) (2003) 99–106.